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Plasma Physics and Controlled Fusion, Vol. 30, No. 12, pp. 1665 to 1689, 1988
Printed in Great Britain.

0741-3335/88 $3.00+ .00
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# NON-MAXWELLIAN ELECTRON DISTRIBUTIONS AND CONTINUUM X-RAY EMISSION IN INVERSE BREMSSTRAHLUNG HEATED PLASMAS

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(*Received* 11 *January* 1988; *and in revised form* 19 *April* 1988)

**Abstract**—The heating of a completely ionized plasma by inverse Bremsstrahlung absorption (IB) is studied by numerically integrating the time varying kinetic equation for electrons, and the resulting distribution functions are used to calculate the continuum X-ray emission. The shape of the distribution functions is determined by the competition between IB and electron–electron collisions, and, in all our uniform plasma simulations, we find that the distributions are well fitted by the formula $C_m \exp[-(v/v_m)^m]$ where $m$ is between 2 and 5. We find the dependence of $m$ on $\alpha = Z^2 v_{osc}/v_{th}^2$, where $Z$ is the atomic number, $v_{osc}$ is the oscillation velocity in the laser field and $v_{th}$ is the thermal velocity: $m(\alpha) = 2+3/(1+1.66/\alpha^{0.724})$. More importantly, in the realistic case of a space and time dependent kinetic simulation of a Be plasma heated by a Nd laser, including heat flow and ion motion, this fit remains valid for electron densities below $0.75N_c$ (with constant $m$), although other effects come into play closer to the critical surface. Thus, these simple results can be used to determine the continuum X-ray spectrum due to Bremsstrahlung (BR) and to direct radiative recombination (DRR). The contributions of BR (valid for average and large atomic numbers) and DRR into each shell are given in a parametric form as functions of $m$ and $T_e$. The implications for the plasma diagnostics based on continuum X-ray emission are discussed.

## 1. INTRODUCTION

COLLISIONAL absorption of laser radiation or inverse Bremsstrahlung (IB) is the main mechanism for the coupling of laser light in laser–target interaction at moderate intensity and/or short wavelength (MOTZ, 1979; DUDERSTADT and MOSES, 1982; MAX, 1980), and is always present at any intensity. Therefore, an accurate characterization of the non-Maxwellian distribution functions produced by IB is essential to the understanding of the physics of the underdense plasma, and the evaluation of continuum X-ray emission from such distributions is necessary to analyse experimental spectra.

While IB heats the bulk of the electrons, primarily the slowest ones, anomalous absorption mechanisms such as resonance absorption or parametric instabilities produce a superthermal tail with a Maxwellian distribution at a higher temperature $T_{hot}$ if the laser intensity is high or the laser wavelength is long. The emission from this component could be easily calculated by treating it as an additional Maxwellian at its own (hot) temperature. Therefore, it will not be discussed further in this article.

When IB heating is very slow, electron–electron collisons enforce a Maxwellian

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distribution. On the contrary, if the IB heating rate is very strong, Langdon's (1980) study in the case of a uniform plasma with no heat flow has shown that the distribution function is severely deformed, taking a form in $\exp[-(v/u(t))^5]$, with $u(t)$ increasing in time as the plasma is heated. In intermediate cases (IB competing with e–e collisions), he plotted the distribution functions for various values of the important parameter $\alpha$ which describes the relative importance of IB heating compared to e–e collisions. It is defined by $\alpha = Zv_{\text{osc}}^2/v_{\text{th}}^2$ where $Z$ is the atomic number, $v_{\text{osc}}$ is the oscillation velocity of an electron in the oscillating electric field and $v_{\text{th}}$ is the thermal velocity. Furthermore, the reduction of absorption due to this deformation of the distribution function was treated and related to $\alpha$ by a simple formula. The problem of inverse Bremsstrahlung was treated more extensively by JONES and LEE (1982), who verified the $\exp[-(v/u(t))^5]$ formula, using three different kinetic equations: Boltzmann, BBGKY as well as the usual Fokker–Planck approach. They also performed Monte Carlo simulations, which showed that this result is valid even when $v_{\text{osc}}^2/v_{\text{th}}^2$ is of the order of one. Another important result is that the anisotropy induced by IB is small: the anisotropic component of the pressure tensor is only $\frac{1}{45}(v_{\text{osc}}^2/v_{\text{th}}^2)$, a result which had been shown previously by MORA and PELLAT (1979). This fact allows one to neglect the induced anisotropy in all moderate intensity simulations.

Whereas the previously cited workers treated a uniform plasma whose energy density rose continuously throughout the calculation, the problems of absorption and transport in a non-uniform plasma were subsequently simulated by ALBRITTON (1983) and later by MATTE *et al.* (1984) who also included the effect of ion motion. In these papers, distribution functions were plotted and discussed, but no attempt to find a simple fit was made.

Recently, MATTHEWS *et al.* (1984) have reported preliminary experimental observations of the deformation of the emission spectra due to the non-Maxwellian character. Using streaked X-ray spectroscopy and blocking the overdense region with a mask, they showed an almost flat spectrum between 2.2 and 3 keV which is consistent with the Langdon electron energy distribution for the low energy part, but is in contrast to the shape expected from a Maxwellian distribution, unless one assumes a very high temperature. As expected, at the end of the laser pulse, the spectrum took a Maxwellian shape during the cooling down phase. These observations are in good agreement with the calculations quoted above, but it would be very desirable—though difficult—to observe also the higher energy part of the spectrum (up to several $kT_e$), where the falloff is steeper than for a Maxwellian.

In a recent paper, we showed (ALATERRE *et al.*, 1986) that such a deformed distribution function also causes a considerable reduction in the excitation and ionization rates in the corona, which further underscores the necessity of a good understanding of the electron energy distribution in the underdense plasma.

The continuum X-ray spectra due to Maxwellian distributions are well known. Recently, LAMOUREUX *et al.* (1984, 1985*a*) have studied the continuum X-ray spectra due to both Bremsstrahlung (BR) and direct radiative recombination (DRR) in the case of a strongly IB heated plasma (form $\exp[-(v/u)^5]$), and pointed out how this affects the usual temperature diagnostic based on the continuum emission.

In this paper, we will show that, in intermediate cases, the distribution functions obtained in uniform plasma simulations are well fitted by the form $\exp[-(v/u)^m]$ (LAMOUREUX *et al.*, 1985*b*, 1987) where $m$ is an increasing function of the quantity $\alpha$

defined above. A relation between $m$ and $\alpha$ will be given [equation (14) of this paper], satisfying the two already known limits: $m(\alpha \to 0) = 2$; $m(\alpha \to \infty) = 5$. It will be seen that this formula also describes quite well the underdense plasma in our non-uniform plasma simulations. The parametric form will be shown to be very useful in the evaluation of continuum X-ray emission and even to perrmit a relatively simple parameterization of the emission due to DRR into each shell and BR (the latter, only for large or average $Z$). The total emission spectrum will be evaluated for two cases.

DUM (1978) has shown that distribution functions of the form $\exp[-(v/v_m)^m]$ also exist in several types of turbulent plasmas (e.g. in the presence of ion acoustic turbulence), and that values of $m$ of 3.6–4 or even 5 could be obtained. The analysis which we make here for the X-ray emission is therefore also relevant to this situation as well as to laser produced plasmas.

In Section 2A, we give some definitions and coefficients necessary to define the $\exp[-(v/u)^m]$ distribution functions for a plasma of a given density and temperature. In Section 2B, we quote the kinetic equation for the uniform plasma case, with IB heating and electron–electron collisions. In Section 2C, we explain the supplementary effects in a non-uniform (one-dimensional, planar) plasma: spatial variation of the laser intensity, ablation and heat flow.

In Section 3A, we show the distribution functions at different laser intensities for uniform plasma simulations and write the $m(\alpha)$ relation. In Section 3B, the non-uniform plasma results are presented for the underdense plasma. In both cases, the distribution functions are compared to the form $\exp[-(v/u)^m]$.

In Section 4, we discuss X-ray continuum emission spectra. We give the general expressions of emissivity coefficients for Bremsstrahlung (BR) and direct radiative recombination (DRR) into each shell. These differ considerably from the widely used coefficients based on Maxwellian distributions and Kramers cross-sections. We then discuss the necessary modifications to the traditional temperature diagnostics based on continuum emission spectra. Next, we calculate the total emission spectrum for two cases: in the first case, a Be plasma, Bremsstrahlung dominates; while in the second, an Al plasma, DRR is stronger. The section concludes by a quantitative analysis of the above mentioned experiment by MATTHEWS *et al.* (1984).

We present our conclusion in Section 5.

## 2. THE ELECTRON KINETIC EQUATION

We will discuss, in this section, the kinetic equation for $f = f_0(v, t)$ or $f_0(x, v, t)$, the isotropic component of the electron distribution function. In the non-uniform plasma simulation, $x$ is the coordinate along the laser beam. Anisotropy of the electron distributions (and of the resulting emission) due to transport are not treated here, but we have also studied these (MATTE *et al.*, 1984; LAMOUREUX *et al.*, 1987; MATTE *et al.*, 1987; MATTE, 1987).

### 2 A. *Definitions*

Density and energy density (and hence temperature) are given by integrals over $f$ (SHKAROFSKY *et al.*, 1966):

$$N_e(x, t) = 4\pi \int_0^\infty f(x, v, t) v^2 \, dv \tag{1}$$

$$\varepsilon(x,t) = \tfrac{3}{2} N_e(x,t) k_B T_e(x,t) = 2\pi M_e \int_0^\infty f(x,v,t) v^4 \, \mathrm{d}v \tag{2}$$

where $k_B$ is Boltzmann's constant. We shall fit the numerically obtained distribution function to functions of the shape $\exp[-(v/v_m)^m]$. We write, following MORA and YAHI (1982):

$$f_m(x,v,t) = C_m \exp[-(v/v_m)^m] \tag{3}$$

with

$$v_m^2 = \frac{3k_B T_e}{M_e} \frac{\Gamma(3/m)}{\Gamma(5/m)} \quad \text{and} \quad C_m = \frac{N_e}{4\pi} \frac{m}{\Gamma(3/m) v_m^3},$$

or, equivalently:

$$f_m(x,v,t) = C_m \exp\left[ -\left( \frac{M_e v^2}{2k_B T_e} a_m \right)^{m/2} \right] \tag{4}$$

with

$$a_m = \frac{2}{3} \frac{\Gamma(5/m)}{\Gamma(3/m)} \quad \text{and} \quad C_m = \frac{N_e}{4\pi} \left( \frac{M_e}{2k_B T_e} \right)^{3/2} \frac{m a_m^{3/2}}{\Gamma(3/m)}.$$

The second form will be used in Section 4, as it is more convenient for X-ray spectra parameterization. Note that, for the Maxwell–Boltzmann distribution ($m = 2$), $v_2^2 = 2k_B T_e / M_e = 2v_{\mathrm{th}}^2$.

2 B. *The electron kinetic equation for a uniform plasma*

The time evolution of a uniform completely stripped plasma is described by the electron kinetic equation:

$$\frac{\partial f}{\partial t}(v,t) = C_{ee}(v,t) + C_{\mathrm{IB}}(v,t). \tag{5}$$

The Fokker–Planck electron–electron collision operator is:

$$C_{ee} = \frac{1}{v^2} \frac{\partial}{\partial v} \left[ v^2 \frac{D_{\|}(v)}{2} \frac{\partial f}{\partial v} + C(v) f \right], \tag{6}$$

where $D_{\|}(v)$ and $C(v)$ are dependent on $f$ (this fact makes the problem non linear), as described by SHKAROFSKY *et al.* (1966).

The classical inverse Bremsstrahlung heating operator is (LANGDON, 1980):

$$C_{IB} = \frac{A v_{osc}^2}{3} \frac{1}{v^2} \frac{\partial}{\partial v} \left( \frac{1}{v} \frac{\partial f}{\partial v} \right) \tag{7}$$

and $v_{osc}$, the oscillation velocity of an electron in the oscillating electric field ($v_{osc} = e E_0 / M_e \omega$), is related to the laser intensity and wavelength by:

$$\tfrac{1}{2} M_e v_{osc}^2 = 21 \text{ eV} \frac{I_0}{10^{14} \text{ W cm}^{-2}} \frac{\lambda_0^2}{(1.06 \, \mu\text{m})^2} \tag{8a}$$

which gives, for the Langdon parameter $\alpha$:

$$\alpha = 0.042 \frac{I_0}{10^{14} \text{ W cm}^{-2}} \frac{\lambda_0^2}{(1.06 \, \mu\text{m})^2} \frac{1}{k_B T_e (\text{keV})} Z \tag{8b}$$

for a thin, tenuous plasma of negligible absorption. $Z$ is the atomic number. We will see, in part C, how equation (8) is affected if the plasma is dense and absorbs considerably.

Equation (6) was obtained by averaging over the fast time scale, i.e. the oscillation period, which is much less than the collision time in practically all cases of interest.

For the Coulomb logarithm, we use the same value for electron–ion collisions as for electron–electron collisions (Book, 1984):

$$\ln \Lambda_{ee} = \ln \Lambda_{ei} = \ln \Lambda = 24 - \ln (N_e^{1/2} T_e^{-1})$$

where $N_e$ is in $\text{cm}^{-3}$ and $T_e$ in eV.

The assumption of a completely stripped plasma may seem exceedingly restrictive. However, to include the phenomena of ionization and recombination would add considerable complication to the kinetic equations and that problem is outside the scope of the present paper. (Preliminary work by us on the ionization of a Maxwellian plasma seems to indicate, however, that the plasma is cooled but the shape of the distribution function is little affected). Furthermore, the ionization cross-section is normally far smaller than the electron–electron collision cross-section, so that the results obtained while neglecting the former can be expected to be valid in most cases.

The energy absorption of the plasma may be obtained by substituting equation (7) into equation (2). One obtains:

$$\frac{\partial \varepsilon}{\partial t} = \frac{4\pi}{3} A Z \frac{N_e}{N_c} v_{osc}^2 f(v = 0) \tag{9}$$

where $N_c$ is the critical density for the laser frequency, that is $M_e \omega_0^2 / 4\pi e^2$. If $f(v)$ is of form (3), then $f(v = 0)$ is simply $C_m$. The ratio $R$ of the reduction of absorption as compared to a Maxwellian ($m = 2$) is thus readily calculated.

As there is no energy "sink" in this problem, the temperature increases rapidly with time. At constant intensity (constant $v_{osc}$), $\alpha$ decreases in time due to the increasing temperature (increasing $v_{th}$).

Decoster and Deck (1985) have found a numerical self-similar solution to

equation (5) in the case where $\alpha$ is constant, that is, the laser intensity increases in time with the temperature.

2 C. *Supplementary effects in a non-uniform plasma*

When we simulate the interaction of a laser beam with a non-uniform plasma, there are three additional effects which come into play: non-uniformity of the oscillating velocity, heat flow into the overdense plasma and cooling of the plasma by expansion.

When the plasma is non-uniform and the maximum density is larger than critical the beam is reflected and both the incident and reflected beams contribute to the heating of the plasma. Furthermore, the reduction in the group velocity of the wave enhances the oscillating electric field (swelling). Thus, in equation (8), $I_0$ must be replaced by the following expression:

$$I_0 \to I(x) = \frac{I_+(x) + I_-(x)}{(1 - N_e(x)/N_c)^{1/2}} \tag{10}$$

where $I_+$ and $I_-$ are respectively the intensities of the incoming and the reflected beam. We assume normal incidence and we write the WKB equation for $I_\pm$ (MOTZ, 1979; ALBRITTON, 1983; MATTE *et al.*, 1984):

$$\frac{\partial I_\pm}{\partial x} = \mp K I_\pm, \quad K = \frac{4\pi}{3}\frac{N_e}{N_c} AZ \frac{8\pi}{c} f(x, v = 0)(1 - N_e(x)/N_c)^{-1/2}. \tag{11}$$

In the numerical calculation, we integrated equation (11) in each computational cell, assuming a linear (or exponential) density variation and a constant ratio $f(x, v = 0)/N_e$ within the cell. Similarly, we averaged (10) within the cell before applying the heating operator (7). This method is analogous to the treatment of collisional absorption by fluid codes which are widely used to simulate laser–target experiments, such as LILAC (GOLDMAN, 1973), FILM (VIRMONT *et al.*, 1978; SHVARTS *et al.*, 1979) and LASNEX (ZIMMERMAN and KRUER, 1975; BAILEY, 1974). The difference is that these codes assume that the distribution is Maxwellian, hence the use of the classical absorption coefficient (JOHNSTON and DAWSON, 1973), which is equation (11) with $f(x, v = 0) = N_e M_e / 2\pi k_B T_e^{3/2}$ and $T_e$ taken to be constant within each computational cell.

This WKB approximation is valid only if the density scale-length and the mean free path are much longer than the laser wavelength. This is quite well verified in our own simulation (MATTE *et al.*, 1984) of a 1.06 $\mu$m laser beam incident on a fully stripped Be plasma. Planar geometry and normal incidence were assumed. After 600 ps, the density scale length was over 100 $\mu$m in the underdense plasma and the mean free path was 20 $\mu$m ($k_B T_e = 2.3$ keV). The absorption was 37%. In all cases where absorption is moderate and planar incidence is assumed, a standing EM wave is formed (WHITE and CHEN, 1974) and the above argument shows that one can take an average over a wavelength. If the plasma is very collisional, this argument does not hold, but then the absorption is very high and the reflected beam is so weak that there is no standing wave, so that equations (10) and (11) are again justified. In this connection, it is also necessary to discuss the singularity in equations (10) and (11),

at $N_e(x) = N_c$. This singularity is integrable and, in our simulation, the $I$ at the critical cell was 3 times that in the very underdense plasma, because of computational cell averaging. A full EM treatment of absorption gives an Airy pattern for the field (WHITE and CHEN, 1974), with no singularity, and a maximum value of $I \sim 15 I_0$, for typical parameters. In the light of these considerations, a full EM treatment of absorption with very fine zoning (as has been done for fluid plasmas), could also be done in a kinetic simulation, but it is doubtful that the physics presented here would be changed substantially.

The effects of transport, the second element to be discussed, are more complicated. The kinetic simulation of heat transport has been discussed previously and we will not elaborate on it here, except in so far as it directly affects the shape of $f_0$ in the underdense plasma. The physical situation is changed qualitatively from the heating problems discussed previously: the absorbed energy is now carried away into the overdense plasma to heat it and drive the ablation, so that the coronal temperature increases only very slowly. The effect that this has on the distribution function in the underdense plasma can only be obtained by numerical simulation, and will be described in Section 3, part B. However, we can note, at this point, that non-local transport tends to deplete the fast electrons in the top part of the heat front, as has been observed in "pure" transport simulations (no heating) of MATTE and VIRMONT (1982) (see Fig. 3, solid line) and LUCIANI *et al.* (1983). More precisely, it is the electrons with high $v_x$ which are depleted, so transport also tends to give a negative $f_2$ as well as a reduced $f_0$ for high energy electrons. This depletion of fast electrons is in addition to that caused by the inverse Bremsstrahlung absorption process, as discussed above. Of course, the departed fast electrons are replaced by a cooler group which is called the "return current", so as to preserve charge neutrality (BELL *et al.*, 1981; MATTE and VIRMONT, 1982).

We now turn to the third effect, plasma expansion, which cools the plasma by transforming thermal energy into kinetic energy and proceed to show how it affects the shape of the distribution function. The operator describing the effect of expansion on the distribution function $F(x, v_x, v_\perp, t)$ is (SHKAROFSKY *et al.*, 1966; BELL, 1983):

$$\frac{\partial F}{\partial t} + V\frac{\partial F}{\partial x} = \left(\frac{\partial V}{\partial t} + V\frac{\partial V}{\partial x}\right)\frac{\partial F}{\partial v_x} + \frac{\partial V}{\partial x} v_x \frac{\partial F}{\partial v_x}$$

where $V(x, t)$ is the fluid velocity and $F(x, v_x, v_\perp, t)$ is the complete velocity distribution ($v = (v_x^2 + v_\perp^2)^{1/2} = (v_x^2 + (v_y^2 + v_z^2))^{1/2}$) of which $f_0(x, v, t)$ is the isotropic component.

Now, we note that the first term on the right is an acceleration term, of the same form as that due to the electric field, $eE/M_e \partial F/\partial v_x$. Therefore, its effect is simply to change the value of the electric field $E$ which is needed to cancel the current (and thus ensure quasineutrality), and it need not be considered. The second one will now concern us. Averaging over angles, we get for the isotropic component (including the second order Legendre expansion term $f_2$)

$$\frac{\partial f_0}{\partial t} = \frac{1}{3}\frac{\partial V}{\partial x}\left[v\frac{\partial f_0}{\partial v} + \frac{1}{5}\frac{1}{v^2}\frac{\partial (v^3 f_2)}{\partial v}\right]. \tag{12}$$

If $f_2 = 0$, we have the self-similar solution

$$f_0(x,v,t) = g\left(x, \frac{v}{w(t)}\right) \quad \text{with} \quad \frac{\mathrm{d}w}{\mathrm{d}t} = -\frac{1}{3}\frac{\partial V}{\partial x} w. \tag{13}$$

This shows that ablation, *per se*, merely cools the plasma without changing the shape of the distribution function, if it is nearly isotropic. However, if the distribution is anisotropic, then, only the $x$ component is cooled. Thus, a flattened distribution function (negative $f_2$) cools less rapidly than an isotropic one. For example, if $f_2/f_0 = -1$, the cooling rate is 40% slower than if $f_2 = 0$. Now, as we have seen in the discussion on transport, the slow and intermediate energy electrons are very nearly isotropic while the fast ones have a flattened distribution with $f_2/f_0 \simeq -1$. Therefore, expansion cools the fast electrons more slowly than the others and, therefore, tends to make the distribution function less depleted at high energy. Our studies of the effects of plasma expansion and heat flow on the shape of $f_0(v)$ and on the value of $f_2(v)$ have shown that heat flow is the dominant effect in cases where radiation is absorbed in the underdense plasma and energy is transported into the overdense plasma by electron thermal conduction. On the other hand, in the case of exploding foil experiments, where an entirely underdense plasma expands while absorbing laser energy [which keeps the temperature nearly uniform and constant (LONDON and ROSEN, 1986)], the effects of expansion which are explained above are quite important (MATTE, 1987).

## 3. SIMULATION RESULTS FOR THE ELECTRON DISTRIBUTION FUNCTION

### 3 A. *Uniform plasma*

We have numerically studied inverse Bremsstrahlung (IB) heating, concurrently with electron–electron collisions, by numerically advancing the kinetic equation (5) in time, with different laser intensities. In our simulations, the initial state was a 0.1 keV Maxwell–Boltzmann distribution and the simulation was stopped when the temperature [as defined by equation (2)] reached 2 keV. During each computation $\alpha = Zv_{\mathrm{osc}}^2/v_{\mathrm{th}}^2$ decreased because $Z$ and $v_{\mathrm{osc}}$ were constant while $v_{\mathrm{th}}$ increased as the plasma was heated. The distribution functions obtained are plotted on Fig. 1. For two of these simulations, we show the distribution functions $f(v)$ and compare to $f_m(v)$ for three values of $m$: $m = 2$ (Maxwell–Boltzmann), $m = 5$ (LANGDON, 1980, IB only) and $m(\alpha)$. The latter is an ansatz which we will explain shortly and is plotted in Fig. 2. The six computations were made with $\alpha$ values which range from the nearly Maxwellian, $\alpha = 0.0106$, $m(\alpha) = 2.066$, to almost pure IB $\alpha = 33.3$, $m(\alpha) = 4.65$. These are the values of $\alpha$ at the end of each computation. In each case, the $f_{m(\alpha)}(v)$ curve is seen to be close to the simulation result $f(v)$, from zero energy up to several times $k_B T_e$, which makes it a good approximation to the simulation results and therefore useful for spectral analysis.

We now explain our $m(\alpha)$ relationship, which is plotted in Fig. 2 and which we may write down as:

$$m(\alpha) = 2 + 3/(1 + 1.66/\alpha^{0.724}). \tag{14}$$

While it is important to know $f_0(v)$ for all energies up to several times $k_B T_e$, the value

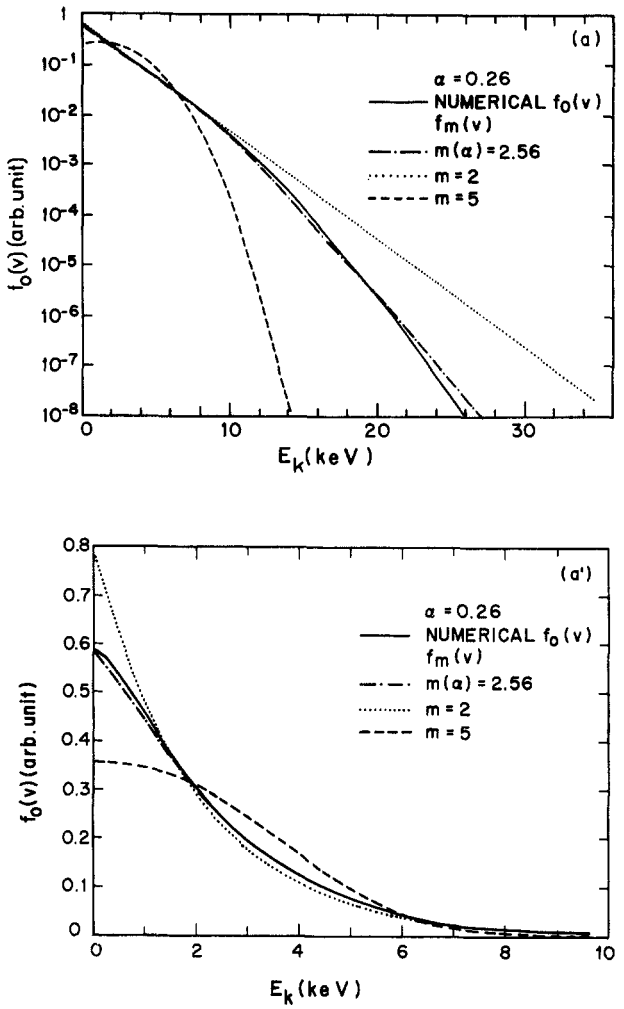



FIG. 1.—Velocity distribution functions, $f_0(v)$ vs $1/2\ M_e v^2$, from uniform plasma simulations at two different intensities. For each, we show also $f_{m=2}(v)$, $f_{m=5}(v)$ and $f_{m=m(\alpha)}(v)$, with $m(\alpha)$ given by equation (14). $\alpha$ and $m(\alpha)$ are indicated on each drawing. The temperature is 2 keV.

at the origin is crucial, since it determines absorption [equation (9)] and LANGDON (1980) has given a formula for the reduction of the absorption—and hence, of $f(v = 0)$—as compared to a Maxwellian:

$$R(\alpha) = f(v = 0)/f_{\mathrm{MB}}(v = 0) = 1 - 0.553/[1 + (0.27/\alpha)]. \tag{15}$$

We have verified that equation (15) is satisfied to within 1% by the use of $f_{m(\alpha)}(v)$ given by formulae (14) and (3). Our own simulation results are very close to equation (14), in both the uniform and non-uniform plasma cases as can be seen in Fig. 2.

A few words are in order, on the behavior of $f_m(v)$. Compared to a Maxwell–

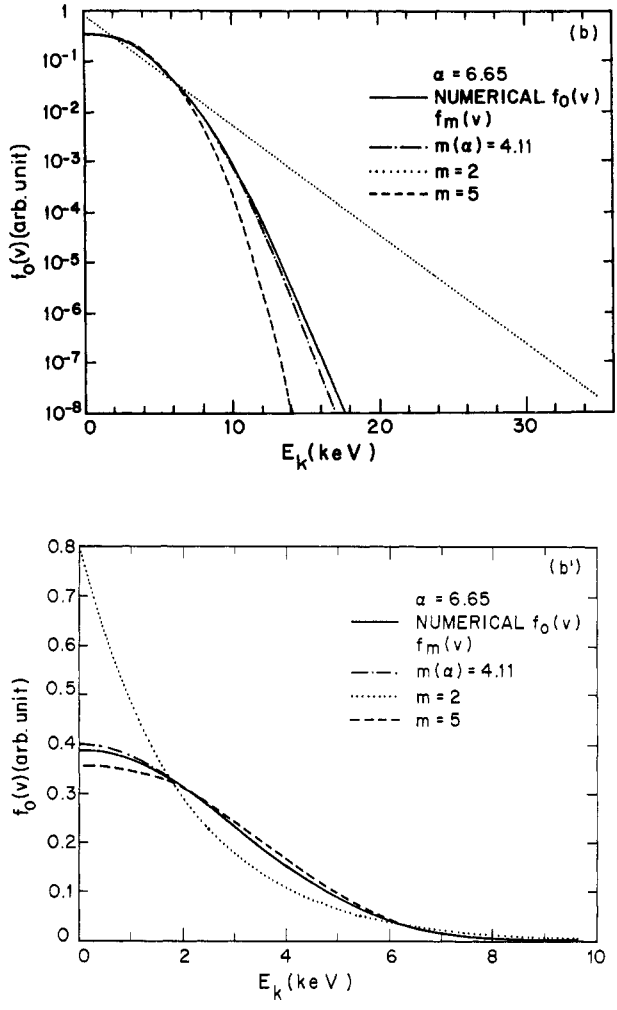



FIG. 1.—*continued.*

Boltzmann distribution at the same density and temperature, the deformed distribution is lower than the Maxwellian between 0 and 1 $k_BT_e$ and above 3 $k_BT_e$, where it drops very rapidly with increasing energy, but is somewhat higher between 1 and 3 $k_BT_e$. Thus, a higher value of $\alpha$ (and hence, of $m$) means fewer slow electrons, far fewer fast electrons, but somewhat more intermediate energy ones. This behaviour is quite easily seen on Figs. 1 and 3.

3 B. *Non-uniform plasma*

We have performed several non-uniform plasma simulations with various laser intensities ($I_0 = 3$–$10 \times 10^{14}$ W cm$^{-2}$), wavelengths ($\lambda_0 = 0.353$ and 1.06 $\mu$m) and atomic numbers ($Z = 4$, 10 and 13). These simulations included advection, electron–electron and electron–ion collisions and an electric field to keep charge neutrality

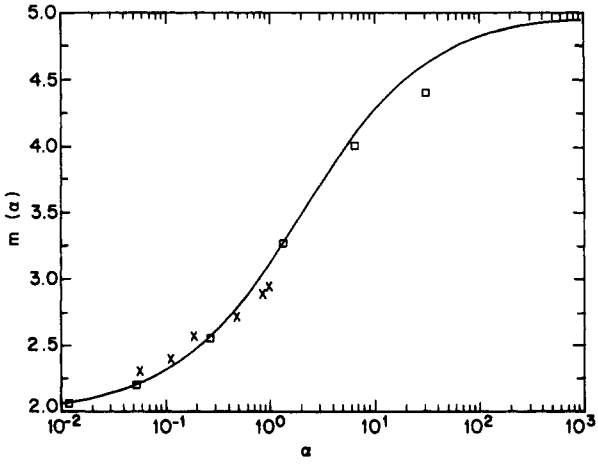



FIG. 2.—Distribution function parameter $m$ vs $\alpha = Zv_{osc}^2/v_{th}^2$, from equation (14). Squares and crosses are least square fits of equation (3) to the numerical distribution functions in uniform and non-uniform plasma simulations respectively.

(MATTE and VIRMONT, 1982), plus inverse Bremsstrahlung absorption and fluid motion. In our best documented case (MATTE *et al.*, 1984), we simulated the interaction of a 1.06 $\mu$m laser beam of intensity $3 \times 10^{14}$ W cm$^{-2}$ on a plasma slab with initial temperature 100 eV and initial density from 0.05 to 20 $\times$ $10^{21}$ cm$^{-3}$ ($10^{21}$ cm$^{-3}$ is the critical density, $N_c$) with scale length of 43 $\mu$m. The atomic number was 4 and total ionization was assumed. After 600 ps, a nearly uniform temperature of 2.3–2.4 keV was seen in the underdense plasma, as well as a density scale length of 120 $\mu$m. As the absorption is only 37%, the total intensity is approximately 1.63 $I_0$ throughout the corona (hence $\alpha = Zv_{osc}^2/v_{th}^2 \simeq 0.46$). However, near the critical surface, the swelling factor is important and $v_{osc}^2$ increased by more than a factor of 2.

It is interesting to note the differences and similarities of the situations without and with heat flow: without heat flow and at this intensity, the temperature rises above 3 keV in less than 20 ps, and would continue to increase rapidly if we allowed it. In contrast, with heat flow and ablation included, we only get 2.3 keV at 600 ps and, by then, the temperature in the corona increases very slowly and the absorbed energy is used to drive the ablation and heat the conduction zone (between the critical surface and the maximum density surface). Nonetheless, if we plot the distribution function and compare it to the uniform density formulas above [equations (3) and (14)] we find very good agreement (better than 5% up to 6 $k_BT_e$, and 7% up to 9 $k_BT_e$) if we look away from the critical surface, for example at $x = 517$ $\mu$m, where the density is 0.40 $N_c$ [Fig. 3(a)]. The value of $m$ (2.77), obtained by a least square fit to the FPI results there, is the same as the value obtained from equation (14). The zoom which we make shows (Fig. 3b) the excellent agreement between the fit and the numerically obtained solution at low energy. This agreement is not totally unexpected for low energy electrons as they "bathe" in a large region (approximately 200 $\mu$) of nearly uniform $\alpha(\simeq 0.5)$. It was pointed out by LANGDON (1980) that the ratio of electron–electron collision time to the transit time is below one for slow electrons, thus ensuring the validity of his formula for the reduction of absorption in realistic conditions.

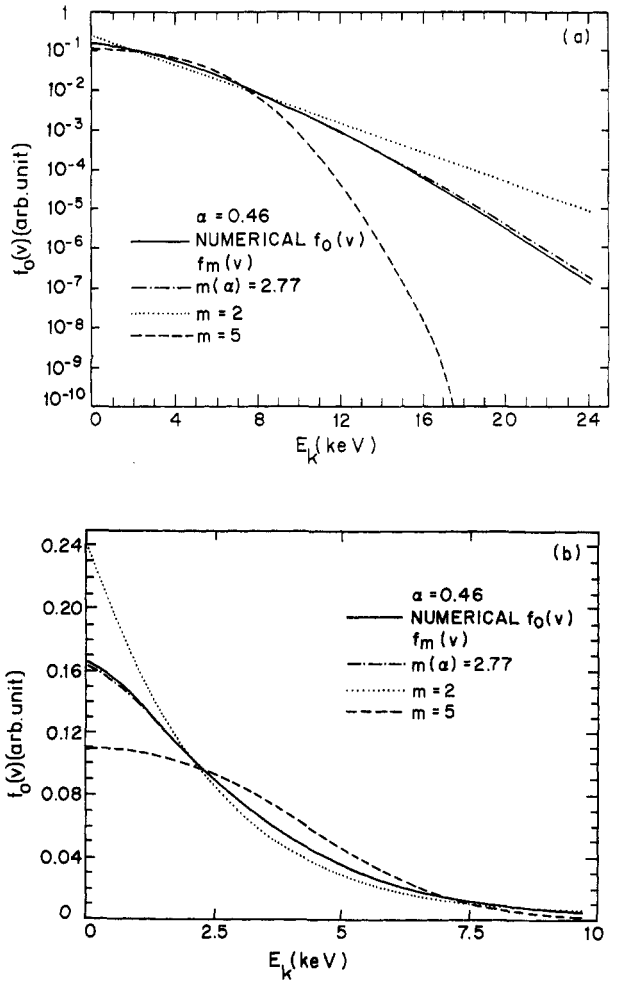



FIG. 3.—Velocity distribution functions, $f_0(v)$ vs $1/2M_e v^2$ from the non-uniform plasma simulation. (a) and (b): far from the critical surface where $N_e = 0.40\ N_c$, $T_e = 2.3$ keV, $\alpha = 0.46$ and $m(\alpha) = 2.77$, also shown are $f_{m=2}(v)$, $f_{m=5}(v)$ and $f_{m=m(\alpha)}(v)$. (c): near the critical surface where $N_e = 0.88\ N_c$, $k_B T_e = 2.3$ keV, $\alpha = 1.1$ and $m(\alpha) = 3.18$; also shown are $f_{m=2.6}(v)$, $f_{m=2.8}(v)$ and $f_{m=m(\alpha)}(v)$. (d) same as (c), but on different scales and $f_m(v)$ curves are with $m = 2.0, 2.8, 3.0, 3.18, 5.0$.

However, this ratio is above one for electrons of energy above $k_B T_e$, the electron–ion collision time is also longer than the transit time. Thus the fact that the higher energy part of the velocity distribution also behaves as in a uniform plasma is a remarkable result. Simulations of non-local transport (MATTE and VIRMONT, 1982; LUCIANI *et al.*, 1983) without heating found some depletion of the high energy part of the distribution in the hot part of the plasma, but in the present case, the main depletion is that due to the absorption process itself. The explanation for this phenomenon is that non-local transport makes the spatial distribution of the high energy electrons far more uniform than the density and temperature profiles would suggest. Indeed,

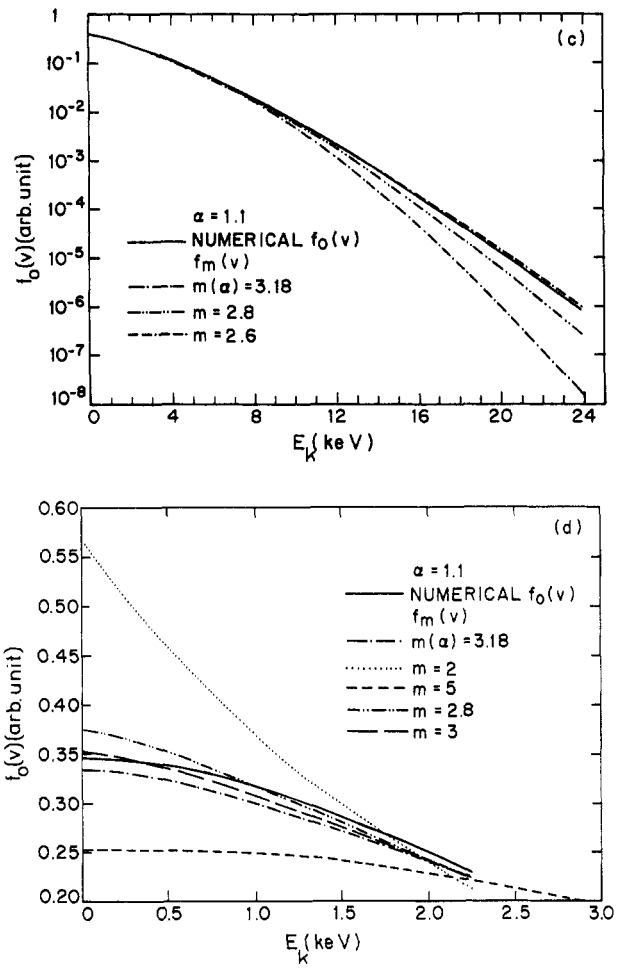



FIG. 3.—*continued.*

after the plasma has absorbed enough energy, the spatial distribution of energetic electrons is seen to extend over several mean free paths into the overdense plasma (MATTE, 1987).

It is important to note that this result was obtained from a simulation using a high order Legendre polynomial ($N = 5$) expansion for the angular distribution. At high energies, the values of $f_2/f_0$ is of order $-1$. In simulations with $N = 1$, the high energy part of $f_0$ was somewhat more depleted than that seen in Fig. (3a) (a least squares fit for $m$ gave $m = 3.1$ instead of 2.77) although the macroscopic quantities (temperature and density profiles, heat flow, etc.) were only slightly different (MATTE *et al.*, 1984). We have found that although $f_2/f_0$ is $\sim -1$ at high velocities, the anisotropic pressure component amounts to only $-2\%$, hence the small effect on the plasma dynamics. Thus, it seems necessary to use a higher order expansion of the angular dependence if one is to simulate spectroscopic diagnostics.

Near the critical surface, however, the situation is different because of two perturbing factors: (1) the rapid variation in space of $v_{osc}^2$, due to the rapid variation of the swelling factor $(1-N_e/N_c)^{-1/2}$ and, (2) the close proximity of the overdense plasmas, where there is no deformation of the electron distribution by the oscillatory field. In Figs. 3(c and d), we show the distribution function at 0.88 $N_c$, where $\alpha = 1.1$ and $m(\alpha) = 3.18$. This point is 11 $\mu$m from the critical surface. In Fig. 3(c), we see the behaviour at all energies. It appears that $m = 2.6$ is the best fit, but $m = 2.8$—which is characteristic of the rest of the underdense plasma where $\alpha \simeq 0.5$ prevails—is a reasonable fit also. The local value, $m(\alpha) = 3.18$ is not as good because fast electrons spend rather little time in this region of very high field. On the other hand, slow electrons are more influenced by the local conditions. In Fig. 3(d), we show a zoom of the distribution function, between 0 and 3 keV, and compare it to the following values of $m$: 2.8, 3.0, $m(\alpha) = 3.18$ as well as 2 and 5. It is seen that $m = 3.0$ is the best fit, but $m = m(\alpha) = 3.18$ is also quite close, though it underestimates the number of slow electrons by a few per cent. The value $m = 2.8$—which is a very good fit below 0.75 $N_c$—would give a slight overestimate of the slow electrons number, and hence of the absorption. The extreme values, $m = 2$ and $m = 5$, are very wrong, by 60 and 30% respectively. Once again, we have emphasized the low energy electrons because of their importance for energy absorption.

Agreement with formulae (3) and (15) was also observed for several other non-uniform plasma simulations, including cases with ultraviolet (0.353 $\mu$m) irradiation. For this good agreement to be obtained, the scale length in the corona must be large enough. From examination of our main simulation ($\lambda_0 = 1.06$ $\mu$m, $Z = 4$, $I = 3 \times 10^{14}$ W cm$^{-2}$, $t = 600$ ps) at earlier times, we noted that, before 200 ps, the scale length was approximately 50 $\mu$m and the corona was far from isothermal, the far corona being 25% cooler than the critical surface, and the high energy part of the distribution function did not yet follow the formulae given above, the fall-off with energy being somewhat less rapid. It is only after 300–400 ps, when the scalelength was above 80 $\mu$m and the coronal temperature was nearly uniform so that the formulae became accurate. For the low energy electrons, on the other hand, agreement was good at all times. At shorter wavelength and high intensity ($\lambda_0 = 0.353$, $I = 10^{15}$ W cm$^{-2}$), the time required to obtain agreement with the formula [in this case $\alpha = 0.18$, $m(\alpha) = 2.44$] was less than 100 ps: the higher absorption meant a rapid build-up of the heat front, with the usual non-local effects of a population of energetic electrons extending several mean free paths into the overdense plasma (MATTE, 1987).

We will now discuss the effect of these non-Maxwellian distribution functions on X-ray spectra.

## 4. CONTINUUM X-RAY EMISSION

The departure of the free electron distributions from the Maxwellian ones will affect to various degrees the atomic quantities generally calculated with this assumption. We study, as an example, the continuum emission due to Bremsstrahlung and direct radiative recombination, versus photon energy. The determination of the emissivity coefficients from the distribution functions and the atomic cross-sections is well known and was recalled in a previous work by LAMOUREUX *et al.* (1984) on emission, restricted to the limiting case $m = 5$ [note that the normalization of the distribution function was $\int f_0(v)v^2\,dv = 1$, instead of $N_e$ as in equation (1) in the present paper]. We will

present results valid for completely stripped ions and will give a few comments on how to extend them to highly charged ions. We will characterize the emission as a function of the parameter $m$ in the distribution function of form (3), $C_m \exp(-(v/v_m)^m)$. We shall take $m = 2.8$, which is representative of the underdense region of the Be plasmas treated in Section 3B. In many circumstances, such as for higher $Z$ and $I$, $m$ would be larger and the non-Maxwellian features then found in the X-ray spectrum would be more pronounced than what is shown below. We have checked that the distribution functions can be used in their analytical form without loss of accuracy as regards the continuum emission. As it is customary to define the individual atomic cross-sections by their Gaunt factors, i.e. by their ratio to the Kramers cross-sections, it will be convenient to express the emissivity coefficients $j$ with reference to the respective standard $j^0$ obtained from Kramers cross-sections in the corresponding Maxwellian plasma at the same temperature and density. We recall that in the latter circumstances the emissivity coefficients for all processes concerned verify:

$$-\frac{\mathrm{d}\ln j(h\nu)}{\mathrm{d}(h\nu)} = \frac{1}{k_B T_e(h\nu)} \equiv \frac{1}{k_B T_e}. \tag{16}$$

This is the basis of a usual spectral temperature diagnostic (LOCHTE–HOLTGREVEN, 1978), which will therefore have to be reconsidered.

4 A. *Bremsstrahlung emission*

When using Kramers' Gaunt factors for highly stripped ions, the emissivity coefficient $j_{\mathrm{Br}}$ for the plasma at the temperature $T_e$ and with the distribution functions $f_m$ as given by (4) are related to the standard $j^0$'s by:

$$\frac{j_{\mathrm{Br}}(h\nu)}{j^0_{\mathrm{Br}}(h\nu)} = \frac{\sqrt{\pi}}{4}\frac{m}{\Gamma(3/m)} a_m^{3/2}\, \mathrm{e}^{h\nu/k_B T_e} \int_{h\nu/kT_e}^{\infty} \exp - \left(\frac{M_e v^2}{2k_B T_e} a_m\right)^{m/2} \mathrm{d}\left(\frac{M_e v^2}{2k_B T_e}\right). \tag{17}$$

The dotted curves of Figs. 4 and 5 show these emissivity results. At low photon energies, the non-Maxwellian character of the plasma has little influence on the emission because the integration is performed over all the electron energies. On the contrary, only the larger ones are involved for high photon energies so that the emissivity results are significantly reduced in comparison to the Maxwellian case, as is illustrated by the dotted curve of Fig. 4. Notice that the choice of the coordinates makes this curve dependent only on $m$, and hence only on $\alpha$.

The consideration of the slope of $\ln j(h\nu)$ according to (16) leads to values of $t_e(h\nu)$ differing from $T_e$ and amounting to:

$$\frac{t_e(h\nu)}{T_e} = \exp\left[\left(\frac{h\nu}{k_B T_e} a_m\right)^{m/2}\right] \int_{h\nu/k_B T_e}^{\infty} \exp\left[-\left(\frac{M_e v^2}{2k_B T_e} a_m\right)^{m/2}\right] \mathrm{d}\left(\frac{M_e v^2}{2k_B T_e}\right). \tag{18}$$

This expression does not lead to an easy analysis of the Bremsstrahlung emission

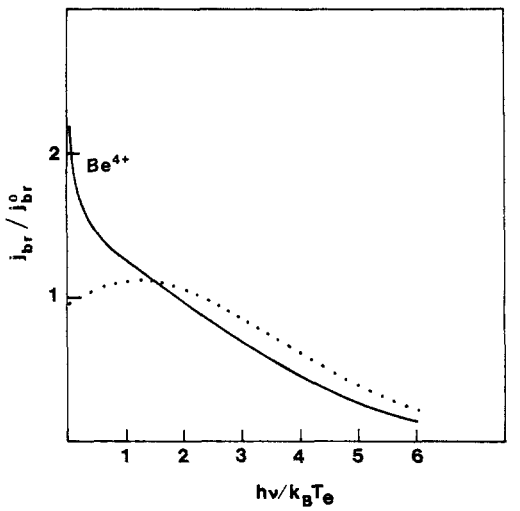



FIG. 4.—The Bremsstrahlung sensitivity coefficients obtained with analytical distribution functions ($m = 2.8$) are compared to those of a Maxwellian plasma ($j_{BR}$) obtained with Kramer's Gaunt factors and plotted vs $h\nu/k_BT_e$. The solid curve (evaluated for the $Be^{4+}$ plasma at $T_e = 2.3$ keV) is obtained with elaborate and the dotted curve with Kramer's Gaunt factors. $h\nu/k_BT_e$ is the ratio of the photon energy to the kinetic temperature.

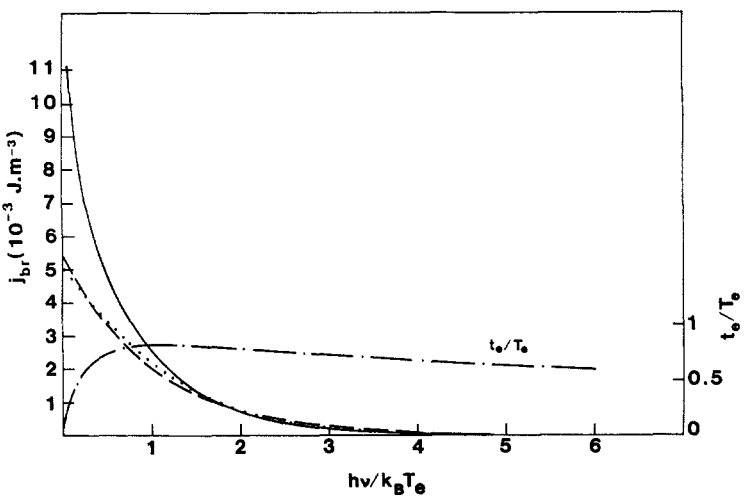



FIG. 5.—Bremsstrahlung emissivity coefficients vs $h\nu/k_BT_e$ for the $Be^{4+}$ plasma at $T_e = 2.3$ keV and $N_e = 0.4\ N_c$. ——: $m = 2.8$ and elaborate Gaunt factors, ····: $m = 2.8$ and Kramers Gaunt factors, – – – –: $j^0_{BR}$, Maxwellian plasma with Kramers Gaunt factors. The dotted-dashed curve, with scale on the right, shows the ratio of $t_e$, deduced by the usual temperature diagnostic, equation (16), (from the solid Bremsstrahlung curve), to the electron temperature $T_e$.

beyond repetitive least square values fittings. When considering two photon energies $h\nu_1$ and $h\nu_2$ in the spectrum, one can establish the more useful expression:

$$\ln \frac{t_e(h\nu_2)\, j_{Br}(h\nu_1)}{t_e(h\nu_1)\, j_{Br}(h\nu_2)} = \left(\frac{a_m}{k_B T_e}\right)^{m/2} [(h\nu_2)^{m/2} - (h\nu_1)^{m/2}]. \tag{19}$$

It could be used either to confirm the validity of the parameterization of the distribution functions or to deduce both $m$ and $T_e$ from an experimental Bremsstrahlung spectrum.

The general expressions (17)–(19) are valid only when making use of Kramers' Gaunt factors, that is Gaunt factors equal to unity. This is justified for completely stripped ions of average and high atomic numbers $Z$ for the temperatures typical of laser plasmas. However, for low $Z$, the Gaunt factors $G(h\nu, v)$ depart significantly from unity, and have to be included as a supplementary factor in the integrand of (17). Using the accurate factors recommended by PRATT and FENG (1984), we found for $Be^{4+}$ the Bremsstrahlung emission represented by the solid curves of Figs. 4 and 5. At low photon energies, it is much larger than the emission estimated from the Kramers Gaunt factors, because the BETHE HEITLER (1934) Gaunt factors then used and whose expression is recalled in the Appendix are larger than unity. In that region of the spectrum, the choice of good quality Gaunt factors is more important than the use of accurate distribution functions. With increasing photon energies, the Kramers data become appropriate and the two estimates reflect the depletion of the tail of the distribution function. Figure 5 shows also the ratio $t_e/T_e$, $t_e$ being deduced from the exactly calculated emission with equation (16). The value of $t_e$ is lower than $T_e$ over the whole spectrum: at low photon energies, owing to the use of accurate Gaunt factors; and at large energies, owing to the non-Maxwellian character of the distribution.

4 B. *Direct radiative recombination*

The emissivity coefficients $j_R$ for recombination into the $n$ shell are directly proportional to the radiative recombination cross-section and to $f_0(v)$, and therefore follow the shape of the distribution function directly. For completely stripped ions, Kramers cross-sections are always satisfactory unlike what we pointed at above for Bremsstrahlung. Use of relativistic exact cross-sections (GOLDBERG, 1982) in $j_R$ modifies the curve of Fig. 6 only slightly in the X-ray range, as we checked for a few sample cases. For atomic systems of ionization degree $Z_i$, it is possible to assimilate the cross-sections to Kramers ones by using the quantity $(Z+Z_i)/2$ (YOUNG SOON KIM and PRATT, 1983). We can also take photoionization cross-sections with a dependence on the photon energy of the form $(h\nu)^\gamma$ where the parameter $\gamma$ depends slightly on the photon energy considered. This has been done for lithium-like ions (BARFIELD, 1979) and should remain valid as long as the cross-sections are monotonically decreasing with $h\nu$, that is as soon as $Z_i$ is larger than a few units (COMBET FARNOUX and LAMOUREUX, 1976). Due to the detailed balance relation, the recombination cross-sections vary as $(h\nu)^{\gamma+2}$. The emissivity coefficients for recombination into the $n$ shell is then:

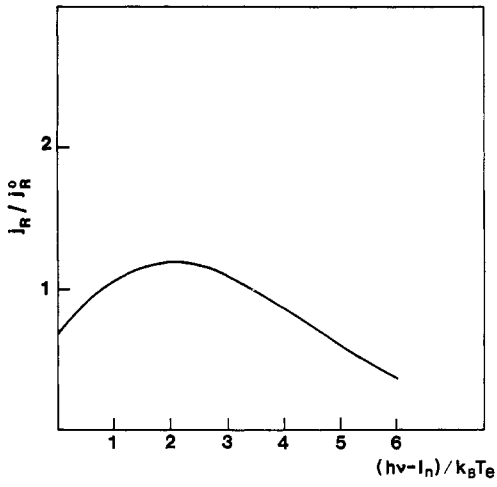



FIG. 6.—The direct radiative recombination emissivity coefficient $j_R^n$ for electron capture on any single shell is calculated with an $m = 2.8$ electron distribution function and compared to $j_R^0$ obtained for a Maxwellian plasma. $j_R^n/j_R^0$ is plotted vs $(h\nu - I_n)/k_B T_e$ where $I_n$ is the recombination threshold of the shell.

$$\frac{j_R(h\nu)}{j_R^0(h\nu)} = (h\nu)^{\gamma+3} \frac{\sqrt{\pi}}{4} \frac{m}{\Gamma(3/m)} a_m^{3/2} \exp\left[\frac{h\nu - I_n}{k_B T_e}\right] \exp\left[-\left(\frac{h\nu - I_n}{k_B T_e} a_m\right)^{m/2}\right]. \quad (20)$$

Once again $j_R^0(h\nu)$ is the emissivity coefficient for a Maxwellian plasma of the same density and temperature. Defining $t_e$ as equation (16), we find:

$$\frac{t_e(h\nu)}{T_e} = \left[-(\gamma+3)\frac{k_B T_e}{h\nu} + \frac{m}{2} a_m^{m/2} \left(\frac{h\nu - I_n}{k_B T_e}\right)^{m/2-1}\right]^{-1}. \quad (21)$$

Results shown in Figs. 6 and 7 correspond to a completely ionized plasma and are thus obtained by taking $\gamma = -3$. The curves $j_R/j_R^0$ and $t_e/T_e$ depend only on $m$. Because of the choice of the abscissa $(h\nu - I_n)/k_B T_e$, these ratios are indeed independent of the atomic number, of the temperature and density, and of the shell into which the electron is recombining. In this example, although the plasma is not extremely non-Maxwellian ($m = 2.8$ is closer to $m = 2$ than to $m = 5$), the departures of these ratios from the unity value are important, but of course less so than for $m = 5$. Notice the typical behaviour of the non-Maxwellian recombination spectrum right at the $I_n$ threshold. The curve $j_R(h\nu)$ in Fig. 7 is comparatively flat at the threshold (this is more obvious when $j$ is plotted on a log scale, see Figs. 8 and 9) so that $t_e$ takes huge values there as illustrated in Fig. 7. As an aid to the analysis of the DRR spectrum and in the same manner as we have done above for the BR spectrum, we now give general relations involving $t_e$ and valid for completely stripped ions:

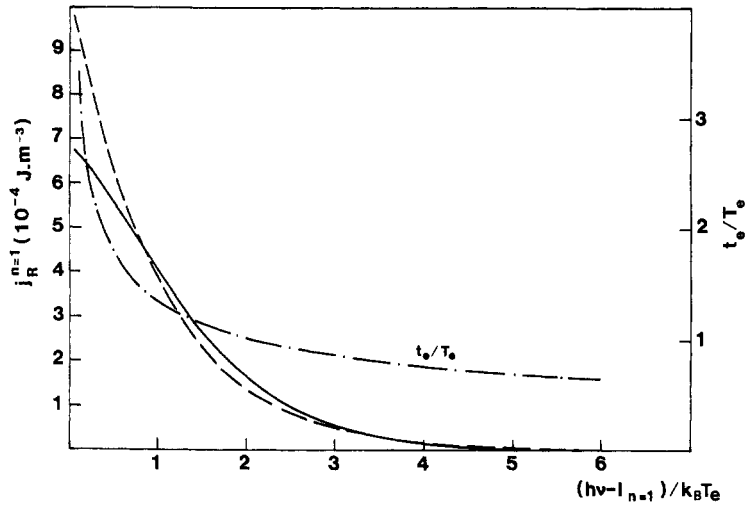



FIG. 7.—Direct radiative recombination emissivity coefficients vs $(hv-I_n)/k_BT_e$. For the $Be^{4+}$ plasma of Fig. 5: ——: $m = 2.8$, – – –: $j_R^0$ for the Maxwellian plasma. The dotted-dashed curve, with scale on the right, shows the ratio of $t_e$, deduced by the usual diagnostic, equation (16) (for recombination into a single shell), to the electron temperature $T_e$.

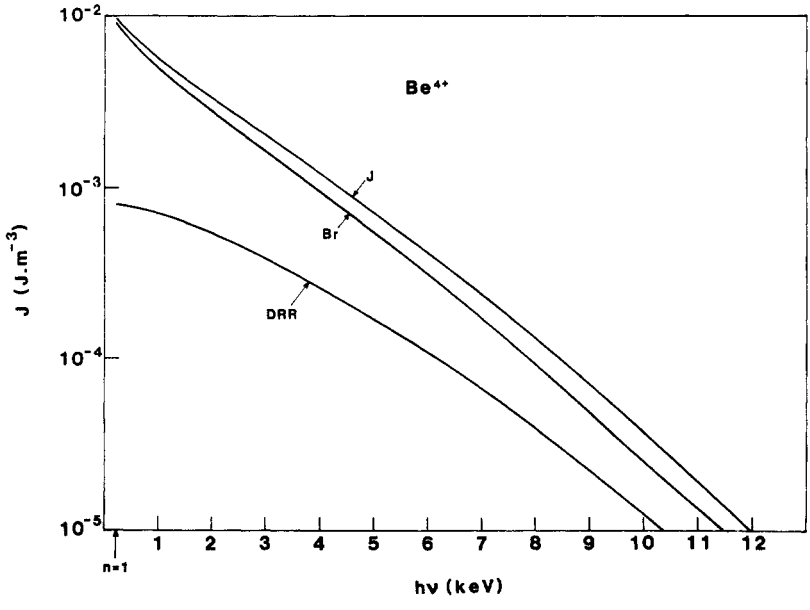



FIG. 8.—Total ($J$), Bremsstrahlung (BR) and direct radiative recombination (DRR) emissivity coefficients vs photon energy $hv$ for the $Be^{4+}$ plasma of Fig. 5 with an $m = 2.8$ electron distribution function. Electron temperature is $T_e = 2.3$ keV and density $N_e = 0.4\ N_c$. The $n = 1$ recombination threshold is indicated by an arrow.

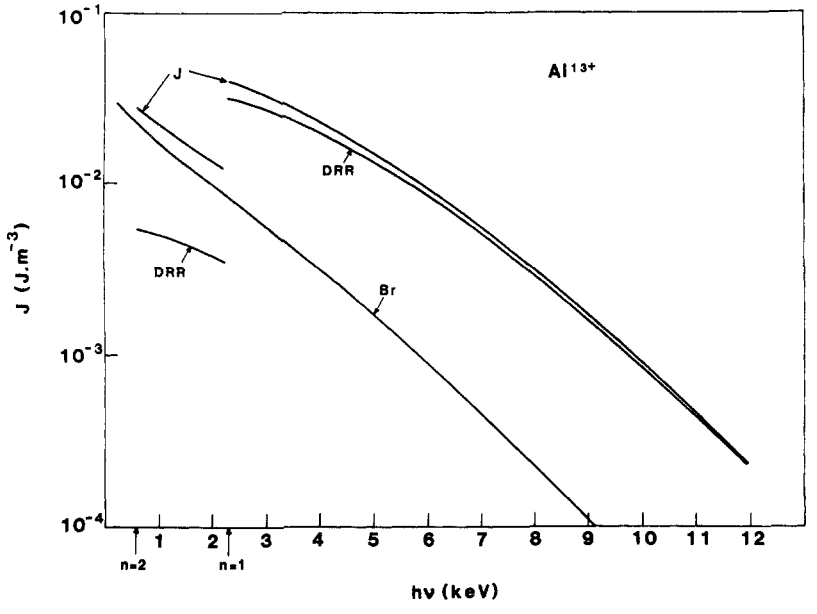



FIG. 9.—Total (Bremsstrahlung plus direct radiative recombination) emissivity coefficients vs photon energy for a totally stripped Al plasma with $m = 2.8$, $T_e = 2$ keV and $N_e = 0.4$ $N_c$. Arrows show the positions of the $n = 1$ and $n = 2$ recombination thresholds.

$$\frac{t_e(h\nu_1)}{t_e(h\nu_2)} = \left[\frac{h\nu_2 - I_n}{h\nu_1 - I_n}\right]^{m/2-1}$$

and

$$-\frac{\mathrm{d}\ln j_R}{\mathrm{d}(h\nu - I_n)^{m/2}} = \left(\frac{a_m}{k_B T_e}\right)^{m/2}. \tag{22}$$

4 C. *Total X-ray emission*

The results given in 4A and 4B show clearly how the emission due to each process is modified by the non-Maxwellian character of the plasma. However, they cannot often be used in an immediate way to predict or interpret a spectrum (even for a plasma consisting of only one completely stripped ion species) since several radiative processes are generally contributing simultaneously. The total emissivity $J(h\nu)$ is obtained by summing over them. Only in the cases of low $Z$ and low photon energies or high $Z$ and high photon energies are BR or DRR into the $n = 1$ shell overwhelmingly dominant, respectively.

For the Be plasma, the total emissivity coefficient $J(h\nu)$ receives its major contribution from the Bremsstrahlung emission. This is illustrated in Fig. 8. The cor-

responding curve $t_e/T_e$ is given in Fig. 10 and shows how misleading it would be to deduce the temperature from the spectrum in the traditional way. The error goes up to 50% which is important, though not to the point of making the mistake obvious. Notice that the diagnostic is invalidated at low photon energies even for Maxwellian plasmas, due to the inappropriateness of Kramers data. A series of calculations were done at various positions in the corona, but not near the critical surface, to check that very little inaccuracy (less than 0.5% for $h\nu \leqslant 2kT_e$ and less than 2% for $h\nu \leqslant 5kT_e$) is introduced by using the $f_{m(\alpha)}$ instead of the numerical distribution functions when evaluating the total emissivity coefficients. More interestingly, the same value of $m$ can be adopted throughout the coronal region of $N_e \leqslant 0.75\ N_c$, owing to the fact emphasized in 3B that $\alpha$ is nearly constant in that region. Indeed, taking $m = 2.8$ for all the plasma at a density below $0.75\ N_c$ adds a mere 0.5% imprecision to the errors mentioned just above.

A similar set of total emission results is given in Fig. 9 for an aluminum plasma with the same value of $m = 2.8$. It shows a spectrum which behaves differently: BR dominates below the $n = 1$ threshold, but DRR strongly dominates above it. This leads to a flat behaviour for $\ln J(h\nu)$ right above the $n = 1$ threshold. Such a behaviour has indeed been observed in space and time resolved spectra (MATTHEWS *et al.*, 1984). As a consequence of it, the value of $t_e/T_e$ shown in Fig. 10 is particularly large just above this $n = 1$ threshold. Notice that the curves $t_e/T_e$ versus $h\nu$ have a completely different behaviour in $Al^{13+}$ and $Be^{4+}$ plasmas, due to the different respective contributions brought to the emission by the BR and DRR processes.

Thus, we see that while both Bremsstrahlung and direct radiative recombination spectra are affected by the non-Maxwellian character of the distribution function, the

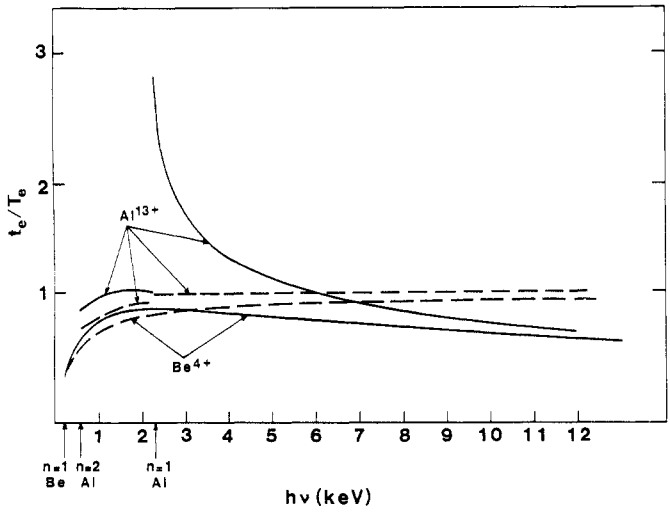



FIG. 10.—Ratio of the temperature $t_e$ deduced by the usual diagnostic, equation (16), (applied to $J$), to the electron temperature $T_e$ vs photon energy $h\nu$, for the Be and Al plasmas of Figs. 8 and 9. Solid curves are for $m = 2.8$ and dashed curves for Maxwellian electron distributions. Arrows show the positions of the $n = 1$ and $n = 2$ recombination thresholds.

latter reflects it even better and therefore a moderate $Z$ target, such as Al, would be more favourable for observing the phenomenon than a low $Z$ material such as Be.

4 D. *Application to the experiment of* MATTHEWS et al. (1984)

We will now apply our results to analyse the experiment of MATTHEWS *et al.* (1984), where an aluminum plasma was produced by a 900 ps pulse of 0.532 $\mu$m light with an irradiance of $5 \times 10^{15}$ W cm$^{-2}$ in a 25 $\mu$m diameter focal spot. The coronal temperature was approximately 1 keV and the recombination spectrum was observed, between 2.2 and 3.0 keV, for the underdense plasma. Emission from the overdense plasma was blocked by a thick mask. Taking formula (20) with $\gamma+3=0$, it is readily seen that the emission at energy $h\nu$ is directly proportional to $f_0(v_v)$ where $h\nu = 1/2\, m_e \times v_v^2 + I$ and $I$ is 2.2 keV in this case. A "flat-topped" spectrum was observed and the authors attributed it to the deformation of the distribution function by inverse Bremsstrahlung heating: in their conditions, they estimated that $\alpha \simeq 6.5$ (actually, the value of $\alpha$ is even higher if the reflected beam is taken into account). Now, even for $\alpha = 6.5$, formula (14) gives $m = 4.1$ and the emission spectrum can be read directly from Fig. 1(b) after shifting the horizontal axis by $I = 2.2$ keV. It is easily seen that the spectrum is indeed quite "flat topped". If we consider the ratio $R = j(h\nu = I)/j(h\pi = I + k_B T_e)$, we get 0.81 for this case while a Maxwellian would give $0.37 \times (1/e)$. It is also interesting to estimate how low the intensity must be so that spectrum is not "flat topped" any more. If we consider $R = 0.61$ $(1/\sqrt{e})$ as a limit, then we find that $m = 2.9$ and $\alpha = 0.6$ whereas if we take $R = 0.5$, we find $m = 2.5$ and $\alpha = 0.22$. Thus, the parameters of the experiment were, to some extent, a case of "overdoing": to see a non-negligible slope in the spectrum, it would have been necessary to reduce the intensity by more than an order of magnitude. This could be done by repeating the same experiment, but with a less tight focus. A spectrometer with a larger energy range would be necessary to observe the rapid decrease which occurs at approximately $2\, k_B T_e$ for large $m$, and this would be a very useful test of the physics exposed here. Note that, because of the very steep gradients which occurred in this experiment (due to the tight focus), we expect that this decrease would be somewhat less rapid than that predicted by the formula, as discussed in Section 3B.

## 5. CONCLUSION

We have determined the isotropic component of the electron velocity distribution in the underdense part of a laser-heated plasma, and calculated the resulting continuum X-ray emission spectrum.

Four competing processes influence the shape of the distribution function: inverse Bremsstrahlung absorption which preferentially heats the slow electrons and tends to give the distribution function an $\exp(-v^5)$ form, electron–electron collisions which tend to restore thermal equilibrium ($\exp(-v^2)$), heat flow which cools the plasma by depleting it of its faster electrons and finally ablation which cools the slow and intermediate energy electrons somewhat more rapidly than the fast ones.

First, several uniform plasma simulations were made, in which only inverse Bremsstrahlung (IB) heating and electron–electron collisions were simultaneously occurring. As the laser intensity was increased, the distribution function went progressively from the Maxwellian ($\exp(-v^2/v_{\text{th}}^2)$) to the pure IB solution, $\exp(-v^5/u^5)$. The distribution is found to depend simply on the Langdon parameter ($\alpha = Zv_{\text{osc}}^2/v_{\text{th}}^2$)

in the following manner: $f(v) = C_{m(\alpha)} \exp(-(v/v_{m(\alpha)})^{m(\alpha)})$ where $m(\alpha) = 2+3/(1+1.66/\alpha^{0.724})$. This formula closely fits the numerically obtained velocity distributions up to energies of several times $k_B T_e$ and also agrees to within 1% with Langdon's formula for the reduction of the absorption, as compared to a Maxwellian plasma of the same density and temperature.

However, the uniform plasma calculations are somewhat artificial as the temperature increases without limit. We have therefore performed kinetic simulations of a laser beam (at $\lambda_0 = 1.06$ and 0.353 $\mu$m) incident on a Be (as well as Ne and Al) plasma, including heat flow and ablation, and found that, in the underdense plasma, the above formula fitted very well the numerically obtained distribution function at densities below 0.75 $N_c$, provided the local laser intensity is correctly calculated (i.e. including swelling, absorption and reflected beam effects). (Between 0.75 $N_c$ and $N_c$, due to the rapid spatial variation of absorption, the fit is slightly less good quantitatively at high energy, though the qualitative features are still valid.) These results imply that one may use the simple uniform plasma time varying results for $m(\alpha)$ and $f(x, v, t) = f_{m(\alpha)}(v)$ for the realistic case of a spatially nonuniform plasma, both for absorption (Langdon effect) and velocity distribution, provided the scalelength is sufficient (at least a few mean free paths).

While the simulations were made in a completely stripped plasma, the distribution function shapes should also be valid in all coronal plasmas where energy loss due to atomic processes (ionization or excitation) in the underdense plasma is small compared to the absorbed energy, that is for moderate $Z$ targets such as carbon and aluminum.

We then used these analytical expressions of $f(v)$ to determine the continuum radiative emission in the underdense region. When Kramers cross-sections are appropriate, this leads to analytical expressions and universal curves depending only on $m$, for the emissivity coefficients $j(h\nu)$ relative to both Bremsstrahlung and direct radiative recombination into a given shell. In such non-Maxwellian plasmas, the quantity $t_e$ defined by $\mathrm{d}\ln j(h\nu)/\mathrm{d}(h\nu) = -1/k_B t_e$ is different from the kinetic temperature $T_e$. We gave relations [equations (18–22)] linking $t_e$, $j(h\nu)$ and $m$ at different photon energies. They could be used in connection with experimental results as a way to confirm the above scaling or to diagnose the non-Maxwellian plasma in view of deducing both $m$ and $T_e$. The influence of the non-Maxwellian character on the total continuum emission was illustrated for two typical cases: Be, dominated by Bremsstrahlung and Al, dominated by recombination. Bremsstrahlung and radiative recombination emission would generally have to be evaluated separately for each plasma situation in which one would be interested.

We finally comment on the practical usefulness of the present work. The simple analytical characterization of the distribution functions enables one to bypass the long numerical determination of the distribution functions as long as estimates are known for the local density, temperature and actual laser intensity (for example from a hydrodynamical code). As we did here for the continuum radiative spectrum, these distributions functions could be used to study other atomic features such as rate coefficients, population determinations, etc., and to investigate further unknown effects due to the non-Maxwellian character of the underdense plasma, in laser–plasma interaction experiments where inverse Bremsstrahlung is the main heating mechanism.

*Acknowledgements*—We thank Dr. C. MOSER and Prof. M. G. HAINES, who organized the CECAM workshop on Laser fusion plasmas which helped further advance this work and the participants for fruitful discussions. We thank Dr. R. H. PRATT for bringing to our attention the work of PRATT and FENG (1984), and Dr. P. JAEGLÉ for helpful discussions. One of us (R.Y.Y.) also wishes to acknowledge support by the Andrew Mellon Foundation, and thanks the CNRS for having enabled him to work in the Laboratoire de Spectroscopie Atomique et Ionique for a few months during the fall of 1984. This research was partially supported by the Natural Sciences and Engineering Council of Canada, the Ministère de l'Education du Québec and the Laser Fusion Feasibility Program at the University of Rochester.

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## APPENDIX

The following expression of the BETHE–HEITLER (1934) Gaunt factors involves quantities expressed in relativistic units ($M_e = h = c = 1$, $\alpha_{FS} = 1/137$, unit of energy = 511 keV). By calling $E_1$, $p_1$, $v_1$ the energy, momentum and velocity of the incident electron, and $E_2$, $p_2$ and $v_2$ the same quantities for the outgoing electron and $k$ the photon energy, the formula reads:

$$G(E_1, k) = \frac{3\sqrt{3}}{16\pi} \frac{p_1 p_2}{E_1^2} A\xi.$$

$A$ is equal to

$$A = \frac{4}{3} - 2E_1E_2\frac{p_1^2+p_2^2}{p_1^2p_2^2} + \frac{\varepsilon_1 E_2}{p_1^3} + \frac{\varepsilon_2 E_1}{p_2^3} - \frac{\varepsilon_1\varepsilon_2}{p_1p_2} + L\left[\frac{8E_1E_2}{3p_1p_2} + \frac{k^2(E_1^2E_2^2+p_1^2p_2^2)}{p_1^3p_2^3} + \frac{k}{2p_1p_2}\left(\frac{E_1E_2+p_1^2}{p_1^3}\varepsilon_1 - \frac{E_1E_2+p_2^2}{p_2^3}\varepsilon_2 + \frac{2kE_1E_2}{p_2^2p_1^2}\right)\right],$$

with

$$L = 2\ln\left(\frac{E_1E_2+p_1p_2-1}{k}\right), \quad \varepsilon_1 = \ln\left(\frac{E_1+p_1}{E_1-p_1}\right), \quad \text{and} \quad \varepsilon_2 = \ln\left(\frac{E_2+p_2}{E_2-p_2}\right).$$

The so-called Elwert factor $\xi$ amounts to:

$$\xi = \frac{v_1}{v_2}\frac{1-\exp(-2\pi Z^*\alpha_{FS}/v_1)}{1-\exp(-2\pi Z^*\alpha_{FS}/v_2)}.$$